BY ORDER OF THE COMMANDER
AIR FORCE RESEARCH LABORATORY
(AFRL)

AIR FORCE RESEARCH LABORATORY
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Scientific/Research and Development

AFRL TEST ACTIVITY INVOLVING HUMAN PARTICIPANTS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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This publication implements AFRL Instructions (AFRLI) 40-402, *Protection of Human Subjects in Research,* and 61-103, *AFRL Research Test Review, Approval, and Oversight*. This instruction provides guidance for review and approval of test activity involving humans (subjects and participants). The intent of this instruction is to ensure AFRL researchers and organizations conducting test activity include appropriate reviews and documentation, assuring that risk for human involvement is minimized. In the event the activity constitutes human subject research, such activity is referred for proper further review as defined by AFRLI 40-402. All applicable civilian personnel policies, instructions, and bargaining agreements will apply. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with (IAW) the Air Force Records Disposition Schedule, which is located in the Air Force Information Management System. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using Department of AF (DAF) Form 847, *Recommendation for Change of Publication*, route DAF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all direct Supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. Submit requests for waivers through the chain of command to the publication OPR in accordance with (IAW) AFRL Delegation of Waiver Approval Authority of Tier Compliance Items Memorandum.

SUMMARY OF CHANGES

This publication has been revised and should be reviewed. Administrative edits and in **Figure 1** changed a block from "Conduct reviews according to guidance in AFRLI 61-103 Volume 2" to "Coordinate with TRB/SRB chairs to ensure appropriate SME participation to cover participant safety concerns" to emphasize needed action. Removed EBS references.

1. Purpose and Scope.

1.1. **Purpose.** The purpose of this instruction is to provide safeguards to enhance human participant and human subject safety in the conduct of research test, evaluation and demonstration activities and to ensure compliance with human research protections IAW AFRLI 40-402. Approval to proceed with human subject testing is provided by the Institutional Review Board (IRB) as applicable in addition to other required approvals such as those mandated by the parent to this volume and Department of the AFI (DAFI) 91-202_AFMC Supplement (SUP)_AFRLSUP, *The US Air Force Mishap Prevention Program*. This instruction provides guidance for obtaining additional reviews as required to meet the purpose stated above.

1.2. **Scope.** The guidance contained in this instruction applies to all AFRL test activities involving human participants/subjects, including experiments and demonstrations that involve AFRL assets (full or part ownership) or AFRL personnel (e.g., government, military, and contractors), or where AFRL either holds mishap accountability or some level of liability. This guidance also applies to all tests executed by organizations under contract to AFRL where AFRL holds either mishap accountability or some level of liability.

1.2.1. Test activities involving humans are defined as those activities which involve an interaction or intervention with an individual. For example, an intervention can involve physical procedures or a manipulation of the individual's environment. An interaction can include such activities as operation of an electromechanical system, medical procedures, psychological testing, surveys, interviews, and focus groups.

1.2.2. Refer to the **Terms** section for further distinction between human participants and human subjects, but a simple characterization is that human participants are not themselves the system under test, but human subjects are.

2. Roles and Responsibilities.

2.1. **AFRL Site/Detachment Qualified Safety Personnel will:**

2.1.1. Assist the Project Manager (PM) in developing the safety component of the test plan and help assess if the activity constitutes human test participant or human subject research.

2.1.1.1. The Human Subject Worksheet (See **Attachment 2**) should be used to aid in making this assessment. This worksheet is to be accomplished for every activity. If the activity does constitute human subject research, it must be reviewed by the AFRL IRB (see AFRLI 40-402).

2.1.1.2. If in doubt, refer the matter to the AFRL IRB for an official and final determination. Only approved staff of the AFRL IRB is authorized to make official human subjects research determinations.

2.1.2. Undergo annual human subjects research training provided by the AFRL IRB. Such training will equip the trainee with the knowledge needed to identify when a test plan needs to be submitted to the AFRL IRB for a human subject research determination or whether or not a test plan constitutes test activity involving human participants. If a test activity involves human participants, it is subject to the relevant stipulations found elsewhere in this Volume (i.e., those found in paragraphs **3.1**-**3.3** and **Attachment 3**).

2.1.3. Confirm that they have taken the required AFRL IRB human subjects research training within one year prior to completing the Human Subject Worksheet (see **Attachment 2**). Contact the AFRL IRB to re-accomplish training if older than one year.

2.2. **Project Manager (PM) will:**

2.2.1. Contact the AFRL Site/Detachment Safety Office and Directorate/Wing Test Lead early in the program development and activity planning stage and provide information concerning test requirements. The PM is responsible for developing a test plan and coordinating that plan with the AFRL Site/Detachment Safety Office and Directorate/Wing Test Lead.

2.2.2. Ensure timely compliance with all applicable requirements and be solely responsible for the planning and execution of the test activity. The PM plans, conducts, manages and documents the activity IAW DAFI 91-202_AFMCSUP_AFRLSUP; AFRLI 61-103; and with this instruction.

2.2.3. Submit test plans to AFRL IRB for official research determination when needed, or when provisions of DoDI 1500.89_DAFI 99-103, *Capabilities-Based Test and Evaluation*, Section 6.4.5 apply to the test activity.

2.3. **Test Execution Authority (TEA) will:**

2.3.1. Accept the residual safety risk as assessed by the safety review process.

2.3.2. Undergo training provided by the AFRL IRB prior to approving a test activity involving human subject research. Such training will equip the TEA with the knowledge needed to understand any stipulations levied by the IRB on a test activity involving human subject research.

2.3.3. If the TEA has an AFRL IRB training date within one year prior to accomplishing TEA responsibilities for a test plan, such training need not be repeated. Contact the AFRL IRB to re-accomplish training if older than one year.

2.4. **AFRL Institutional Review Board (IRB) will:**

2.4.1. Provide annual human subjects research training to AFRL safety personnel when contacted by the Site/Detachment Safety Office.

2.4.2. Provide annual human subjects research training to TEA when contacted by the TEA.

3. Test Process.

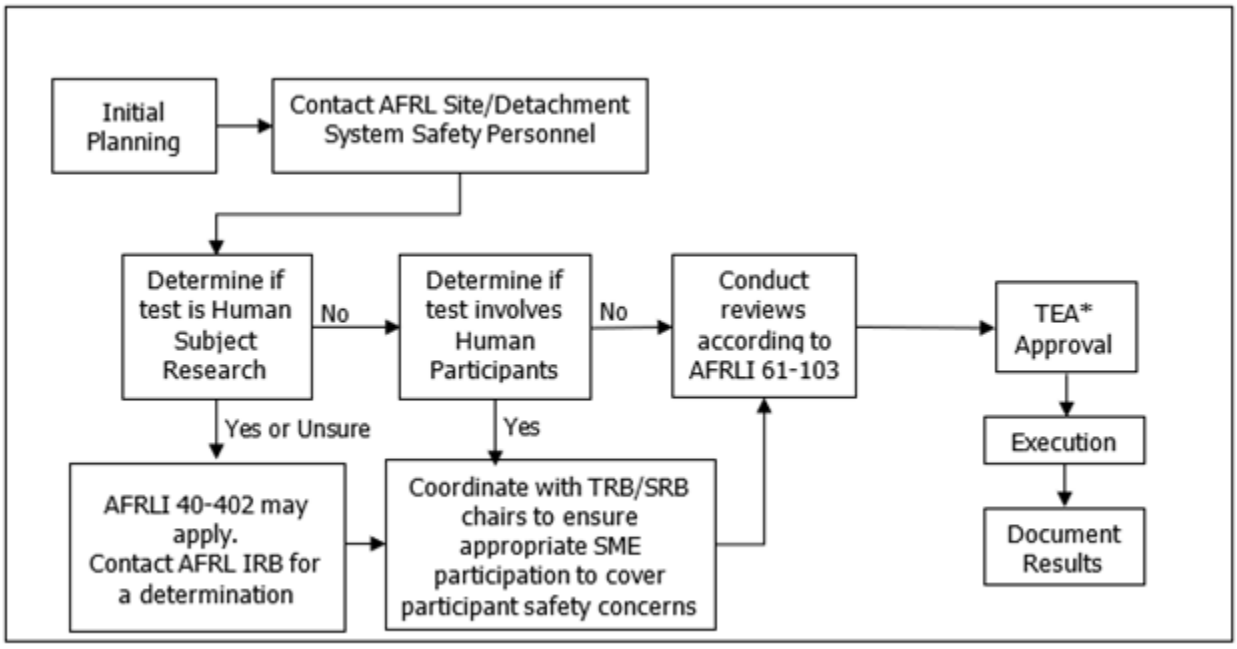
3.1. **AFRL Test Activity.** All AFRL test activity which potentially involves human participants, or human subjects will follow the determination and review process as depicted in **Figure 1**. When test activities include human subjects or human participants, items related

to their participation must be listed in the test plan as described in **Attachment 3**. IAW DAFI 91-202_AFMCSUP_AFRLSUP all test plans must be coordinated with the appropriate qualified professional at the AFRL Site/Detachment Safety Office prior to conducting any tests with human participants and the AFRL IRB if involving human subjects.

3.2. **Testing.** The test should be conducted in such a way as to avoid any mental or physical harm or unnecessary discomfort to participants. The test plan must detail the safety aspects taken into account for the test. During the course of the test, the PM, scientist, or engineer in charge must be prepared to terminate the test at any stage, if he or she has any cause to believe, in the exercise of good faith, superior skill and careful judgment that continuation of the test is likely to result in injury, disability, or death to the participant, or is unlikely to produce the expected outcome through technical, Risk Management (RM), safety, or other analysis method. No individual under the age of 18 will be a test participant. Test participants must voluntarily provide informed consent to being a test participant. Any individual unable to provide consent will not be used as a test participant.

3.3. **Human Participant Data.** All information about a human test participant will be maintained consistent with existing Privacy Act and DoD Privacy Rules, and in accordance with records management and Information Assurance requirements as applicable.

Figure 1. Research Test Management Process for Human Participants.



* Flight Operations Authority approval also required for flight test activities.

JASON BARTOLOMEI, Major General, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

Title 32 CFR Part 219, *Protection of Human Subjects*, current edition

DoDI 3216.02_DAFI 40-402, *Protection of Human Subjects and Adherence to Ethical Standards in DAF-Conducted and -Supported Research*, 17 January 2024

DoDI 5000.89_DAFI 99-103, *Capabilities-Based Test and Evaluation*, 9 December 2021

AFI 33-322, *Records Management and Information Governance Program*, 28 July 2021

DAFI 91-202_AFMCSUP_AFRLSUP, *The US Air Force Mishap Prevention Program*, 11 October 2023

AFRLI 40-402, *Protection of Human Subjects in Research,* 11 March 2025

AFRLI 61-103, *AFRL Research Test Review, Approval and Oversight,* 26 September 2025

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AFRL IRB—Air Force Research Laboratory Institutional Review Board

AFRLI—AFRL Instruction

AFI—Air Force Instruction

CFR—Code of Federal Regulation

DAF—Department of Air Force

DAFI—Department of Air Force Instruction

DoDI—Department of Defense Instruction

IAW—In Accordance With

OPR—Office of Primary Responsibility

RM—Risk Management

PM—Program Manager

RDT&E—Research, Development, Test and Evaluation

TEA—Test Execution Authority

TD—Technology Directorate

TRB—Technical Review Board

SRB—Safety Review Board

USC—United States Code

Office Symbols

711 HPW/EN—711 Human Performance Wing, Engineering

711 HPW/DV—711 Human Performance Wing, Vice Director

711 HPW/IR—711 Human Performance Wing, Institutional Research

Terms

Generalizable Knowledge—New information that has relevance beyond the population or program from which it was collected, or information that is added to the scientific literature.

Human Subject Research—A systematic investigation, including research development, testing and evaluation, designed to develop or contribute to generalizable knowledge, when there is intervention or interaction with a human as the subject of the research. This does not include activities, including program evaluation, customer satisfaction surveys, user surveys, outcome reviews, and other methods, designed solely to assess the performance of DoD programs where the results of the evaluation are only for the use of Government officials responsible for the operation or oversight of the program being evaluated and are not intended for generalized use beyond such program. (For further discussion, guidance, and related terminology see DoDI 3216.02_DAFI 40-402, *Protection of Human Subjects and Adherence to Ethical Standards in DAF-Conducted and -Supported Research*, and AFRLI 40-402, *Protection of Human Subjects in Research).*

Human Subject—A living individual about whom an investigator conducting research obtains:

(1) ——Data through intervention or interaction with the individual, or

(2) ——Identifiable Private Information (PII).

(3)—(For further discussion, guidance, and related terminology see DoDI 3216.02_AFI 40-402).

Human Participant—A living individual who is not the subject of the test activity, but who affects the test activity through intervention or interaction. The focus of data collected is not about the individual but rather is about the object or design being tested.

Qualified Safety Person—Defined by AFI 91-202 as **"**Qualified Safety, Fire Protection, Bioenvironmental Engineering and Health Officials—Air Force civilian and military personnel assigned to full-time positions for the respective disciplines. Air Force civilian personnel who meet the Office of Personnel Management standards for safety and occupational health manager or specialist, safety engineering technician, safety engineer, fire protection engineer or specialist, medical officer, health physicist, industrial hygienist, occupational health nurse or environmental health technician job qualification standards. Safety, fire protection, and health personnel with experience or up-to-date training in occupational safety, fire protection, and health hazard recognition and evaluation are considered as meeting the qualifications of safety, fire protection, and health inspectors. Air Force military personnel, who possess a safety, fire protection, Bioenvironmental Engineering, aerospace medicine, or medicine Air Force Specialty Code (AFSC). Government civilian employees are considered fully qualified IAW 29 CFR Part 1960 and military at the AFSC 7-level or can be a 5-level if task certified. (See 29 CFR § 1960.25).

Test Activity—For the purposes of this instruction, is an evaluation, demonstration, development, experiment, or other effort, which may or may not encompass Human Subject Research.

Test Activity Involving Human Participants—Any test activity that includes the use of humans through intervention or interaction and is NOT Human Subject Research. This typically applies in cases where the human being is not the focus of the test activity. For example, to use a human to evaluate the sound quality and physical comfort of a stereo headset, the human would not be the focus/subject of the test.

Attachment 2

WORKSHEET FOR ASSESSING APPLICABILITY OF HUMAN SUBJECTS RESEARCH REGULATIONS - EXAMPLE

This worksheet is to be accomplished for each AFRL activity.
Step 1: Does this project meet the definition of research, per the regulatory definition found in 32 CFR 219:
Research is defined as: A **systematic** investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge. Activities that meet this definition constitute research whether or not they are conducted or supported under a program that is considered research for other purposes.
_____A. *The activity is a **systematic** investigation*. A **systematic investigation**: involves a predetermined, organized plan for collecting information. Utilizes research methodology and includes a plan for data analysis.
_____B. *Designed to develop or contribute to **generalizable** knowledge:* Includes activities that may not produce generalizable knowledge in of themselves but are intended to lead to it (for example: preliminary or pilot studies as the basis for further research). Is designed to yield answers to a clearly stated research question, test a specific hypothesis, or develop a theory. **Generalizable knowledge:** The intent to expand understanding of a condition or population or add to the body of knowledge regarding a field of study. Generalizable knowledge means new information that has relevance beyond the population or program from which it was collected, or information that is added to the scientific literature.
Note: This does not include activities, including program evaluation, customer satisfaction surveys, user surveys, outcome reviews, and other methods, designed solely to assess the performance of DoD programs where the results of the evaluation are only for the use of Government officials responsible for the operation or oversight of the program being evaluated and are not intended for generalized use beyond such program.
If YES to BOTH A and B above, the project meets the definition of **research.** Continue below to step 2 to determine if it qualifies as HUMAN SUBJECTS RESEARCH:
Note: If a project has multiple components and at least one of those components is considered human subjects research, the entire project is classified as human subjects research unless the components are separable.
Step 2. Does this project include HUMAN SUBJECTS per the regulatory definition found in 32 CFR 219?

_____C. This project includes obtaining information **or** bio specimens through *intervention* or *interaction* with the individual and uses, studies or analyzes the information or bio specimens. **Intervention** includes both physical procedures by which information or bio specimens are gathered (such as venipuncture) and manipulations of the subject or the subject's environment that are performed for research purposes. **Interaction** includes communication or interpersonal contact between investigator and subject.
_____D. This activity will obtain, use, study analyze or generate identifiable private information or identifiable bio specimens. **Identifiable private information** is private information for which the identity of the subject is or may readily be ascertained by the investigator or associated with the information. **Identifiable bio specimen is a bio specimen** for which the identity of the subject is or may readily be ascertained by the investigator or associated with the bio specimen.
If YES to C **OR** D, the research involves human subjects. To sum, if YES to A **AND** B **AND** either C
OR D, the activity is subject to human subjects research regulations per this worksheet. Indicate below:
_____Yes ____No ____Indeterminate/Not Sure
Step 3: Does this project involve FDA-regulated investigational device activity, per 21 USC 321 and 812:
_____ Is a "medical device" (e.g., a device that effects the function or structure of the human) the focus of the activity, and a human is being used to assess the safety or effectiveness of the device?
If the activity is either FDA or human subjects research regulated activity (or Indeterminate/Not Sure) per this worksheet, the project manager must ensure the activity is reviewed by the Air Force Research Laboratory, Institutional Research (711th HPW/IR) IAW all applicable Federal, DoD and local regulations. The AFRL IR administrative staff can be reached at 937-904-8100 or afrl.ir.protocolmanagement.@us.af.mil Templates: https://usaf.dps.mil/teams/10213.

Attachment 3

ITEMS TO INCLUDE IN A TEST PLAN INVOLVING HUMAN PARTICIPANT IN TEST ACTIVITIES

A3.1. If living humans are required to participate in the test activity outside the role of the Program Manager, investigator, or investigative staff, of a test, the following items must be included in this section: *NOTE:* This outline contains many suggested topics to trigger your thoughts to cover all necessary topics. However, some may not be applicable to your specific program. Tailor the outline to fit your program.

A3.1.1. Total number of participants

A3.1.2. Inclusion/exclusion criteria: screening, qualifications (skills) for participants and/or special tests required

A3.1.3. Identify participant's background (active/retired military, reserve, civilian)

A3.1.4. Age range of participants

A3.1.5. Risk factors and mitigation of risk efforts

A3.2. Describe in detail the role the human participants will play in the test. For example, what a participant will experience while taking part in the test. Summarize all pertinent information needed to follow the complete course of a test session and perform an adequate evaluation of the test design with regard to human participation. This section should lay out in detail how the participant will interact with test device or equipment, whether directly or indirectly.

A3.3. If any special skills are required of participants or investigative staff, the ability of participants or investigative staff to accomplish those skills must be evaluated prior to test participation (examples include swimming, sky diving, serving as a pilot, driving, licensure, etc.).

A3.4. Describe all possible hazards, risks and discomforts to participants to include both physical and psychological risks. For each identifiable risk, provide information on its incidence, the availability and effectiveness of treatment, and possible long-term or permanent effects. Include information on efforts taken to minimize any risks and maximize safety. Provide a comprehensive summary of all medical support requirements of the test program to include precautionary or preventative medicine measures required, if applicable.

A3.5. Specify any support required or any special safety precautions that need to be in place to ensure the protection of the human participant. If medical observers are needed, clearly state their qualifications, roles, and responsibilities.